

Program : Diploma in Cyber Forensics and Information Security	
Course Code : 6281C	Course Title: Mobile Application Security
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L: 4 T: 0 P: 0)	Periods per semester: 60

Course Objectives:

- Understand the Android application threat landscape
- Secure mobile devices, their data, and applications

Course Prerequisites:

Topic	Course code	Course name	Semester
Common security concepts	3281	Fundamentals of Cryptography and Information Security	III
Basic knowledge of the Linux OS	3157	System administration lab	III

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Introduction to android architecture & Security	15	Understanding
CO2	Understand the Android application threat landscape	15	Applying
CO3	Introduction to ios architecture & Security	15	Applying
CO4	Understand the iOS application threat landscape	15	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	2						3
CO2	1	2		1			2
CO3	2	2			1		
CO4	2			1			3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Introduction to android architecture & Security		
M1.01	Android architecture, Android virtual machine, Android security model	5	Understanding
M1.02	Setting up a testing environment	3	Understanding
M1.03	Android build process, Reversing APK	4	Understanding
M1.04	Device Rooting	3	Applying
Contents: Android architecture, Android virtual machine, Android security model, Setting up a testing environment, Android build process, Reversing APK Device Rooting,			
CO2	Understand the Android application threat landscape		
M2.01	Network Traffic, Device and data security	5	Understanding
M2.02	Tap Jacking	4	Understanding
M2.03	Static code Analysis	3	Applying
M2.04	Dynamic code analysis	3	Applying
	Series Test – I	1	
Contents : Network Traffic, Device and data security, Tap Jacking, Static code Analysis, Dynamic code analysis			

CO3	Introduction to ios architecture & Security		
M3.01	Ios Architecture	5	Applying
M3.02	Device Jail breaking	4	Applying
M3.03	Setting up a testing environment	3	Applying
M3.04	ios building process	3	Applying
Contents : Ios Architecture, Device Jail breaking, Setting up a testing environment, ios building process			
CO4	Summarize the working and applications of diodes, transistors and logic gates		
M4.01	Reversing ios applications	4	Applying
M4.02	ios application fundamentals, ios testing fundamentals	3	Applying
M4.03	Network Traffic	5	Applying
M4.04	Device administrator, Dynamic Analysis	3	Applying
	Series Test – II	1	
Contents : Reversing ios applications, ios application fundamentals ,ios testing fundamentals, Network Traffic, Device administrator, Dynamic Analysis			

Text / Reference

T/R	Book Title/Author
T1	Mobile Application Security by David Thiel, Chris Clark, Himanshu Dwivedi
T2	Mobile Application Security McGraw-Hill, Inc. Professional Book Group 11 West 19th Street New York, NY

Online Resources

Sl.No	Website Link
1	https://www.elearnsecurity.com/maspt